

# Deep Neural Networks - Mid-Semester 1 Mock 01 Solutions

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# DNN Midsem 1 Mock 01 - Answer Key

## Q1

(a) Output 1 when  $z \geq 0$ , otherwise 0. Finite convergence requires a finite, linearly separable training set repeatedly processed by the standard mistake-driven perceptron procedure with positive learning rate; no such promise covers XOR/nonseparable data.

(b) Error  $y - \hat{y} = 1$ .  $w_{\text{new}} = (0.5, -0.2) + 0.2(1)(2, 3) = (0.9, 0.4)$ .  $b_{\text{new}} = -0.1 + 0.2(1) = 0.1$ . Both columns have shape  $2 \times 1$ ; the update moves the score upward for this positive example.

## Q2

(a) Width is units within a layer; depth is successive computational transformations under a declared counting convention. If  $h = W_1x + b_1$  and  $y = W_2h + b_2$ , then  $y = (W_2W_1)x + (W_2b_1 + b_2)$ , still one affine map. Nonlinear activations are required to avoid this collapse.

(b)  $W_1: 4 \times 5$ ,  $b_1: 4 \times 1$ ,  $W_2: 2 \times 4$ ,  $b_2: 2 \times 1$ . Parameters:  $20 + 4 + 8 + 2 = 34$ . Counting only trainable transformations gives two layers; the input placeholder is excluded.

## Q3

(a)  $z = 2(2) - 1 = 3$ ,  $h = 3$ ,  $L = (3 - 6)^2 = 9$ .  $dL/dh = 2(h - 6) = -6$ ,  $dh/dz = 1$  because  $z > 0$ , and  $dz/dx = 2$ ; therefore  $dL/dx = (-6)(1)(2) = -12$ . All quantities are scalar.

(b) Subtract maximum 3: shifted logits  $(0, -2, -3)$ . Exponentials are approximately  $(1, 0.1353, 0.0498)$ , sum 1.1851, probabilities  $(0.844, 0.114, 0.042)$ . With class 2 correct,  $L = -\ln(0.114) = 2.170$  approximately. Each probability is in  $(0, 1)$  and the vector sums to 1; the fairly large loss reflects low assigned probability for the correct class.

## Q4

(a) A saddle curves upward in some directions and downward in others; a plateau is a broad region with very small slope and may lack opposite curvature signs.

(b) Nesterov evaluates the gradient after a momentum look-ahead and uses velocity history; RMSprop scales the current gradient using an exponential average of past squared gradients.

(c)  $m_1 = 0.9(0) + 0.1(-2) = -0.2$ ;  $v_1 = 0.99(0) + 0.01(4) = 0.04$ ;  $\hat{m}_1 = -0.2 / (1 - 0.9) = -2$ ;  $\hat{v}_1 = 0.04 / (1 - 0.99) = 4$ . Thus  $\theta_1 - \theta_0 = -0.01(-2) / (\sqrt{4}) = +0.01$ . Bias correction offsets zero-initialization bias in early moments. Epsilon must be positive in an implementation, though declared negligible here.

## Q5

(a) Validation data selects checkpoints/hyperparameters; the untouched test data estimates final selected-model performance. Repeated test use contaminates that estimate.

(b) Training improves throughout, while validation improves through checkpoint 3 then worsens: overfitting after checkpoint 3. Select checkpoint 3 by minimum validation loss 0.39. At checkpoint 4, the train-validation gap is  $0.55 - 0.12 = 0.43$  loss units.

(c) Accept early stopping at checkpoint 3, stronger L1/L2/dropout, or more representative data, if linked to validation. A limitation must be stated—for example, regularization can increase underfitting and BN is not a universal cure for distribution shift.

## Q6

(a) Effective height kernel  $2(3-1)+1=5$ ; width kernel  $1(3-1)+1=3$ . Height  $\text{floor}((25+2^2-5)/2)+1=13$ ; width  $\text{floor}((19+2^2-3)/2)+1=10$ . Output shape  $13 \times 10 \times 7$ . Parameters  $3^3 \cdot 4 \cdot 7 + 7 = 259$ .

(b) Dilation spreads height taps so the effective height is 5; stride 2 samples every second output position in both axes. Fixed shared kernels are reused at all positions, so input height/width affect the number of activations, not the learned parameter count. Conditions: positive integer strides/dilations and nonnegative output numerators.

## Q7

Begin with the pretrained feature extractor frozen and train a new target head; select using target validation data. Then unfreeze later blocks gradually with a smaller learning rate and stop if target validation worsens. Thermal versus colour imagery is a domain mismatch that can limit feature transfer; compare against a target-trained baseline and track the train-validation gap. If a residual block halves height/width and doubles channels, its shortcut needs a stride-2 projection producing the same doubled channel count before addition.

## Q8

(a) Universal-approximation capacity does not guarantee that optimization finds parameters or that they generalize. Dropout targets generalization but cannot by itself repair saturated activations/vanishing gradients; high errors on both splits do not show classic overfitting.

(b) A supported diagnosis is vanishing gradients from sigmoid saturation or scale-poor initialization. Inspect activation distributions and layerwise gradient norms, then test a suitable initialization and nonsaturating hidden activation while holding data/splits fixed. Recovery of early-layer gradients with falling training loss would support the diagnosis; validation must still be checked separately.

# DNN Midsem 1 Mock 01 - Point Rubric and Blueprint

Award each point independently. Valid alternative methods earn the same competency marks. Arithmetic follow-through applies after an isolated arithmetic error if later work remains internally consistent and valid.

| Point | Mark | Difficulty | Session | Primary form | Objective demand                                             |
|-------|------|------------|---------|--------------|--------------------------------------------------------------|
| Q1-P1 | 1    | Easy       | 1       | Property     | Exact $z \geq 0$ threshold convention                        |
| Q1-P2 | 1    | Standard   | 1       | Property     | All finite-convergence conditions and nonseparable exclusion |
| Q1-P3 | 1    | Easy       | 1       | Numerical    | Correct vector update with substitution                      |

| Point | Mark | Difficulty | Session | Primary form | Objective demand                                       |
|-------|------|------------|---------|--------------|--------------------------------------------------------|
| Q1-P4 | 1    | Standard   | 1       | Numerical    | Correct bias update, shapes, and interpretation        |
| Q2-P1 | 1    | Easy       | 2       | Comparison   | Correct width/depth distinction and convention         |
| Q2-P2 | 1    | Stretch    | 2       | Comparison   | Algebraic affine-collapse explanation                  |
| Q2-P3 | 1    | Standard   | 2       | Numerical    | All four parameter shapes                              |
| Q2-P4 | 1    | Standard   | 2       | Numerical    | Correct parameter total and stated counting convention |
| Q3-P1 | 1    | Easy       | 3       | Numerical    | Correct forward values                                 |
| Q3-P2 | 1    | Standard   | 3       | Numerical    | Labelled local derivatives and chain product           |
| Q3-P3 | 1    | Standard   | 3       | Numerical    | Stable softmax steps and probabilities                 |
| Q3-P4 | 1    | Stretch    | 3       | Numerical    | Cross-entropy and brief interpretation                 |
| Q4-P1 | 1    | Easy       | 4       | Property     | Saddle/plateau conditional distinction                 |

| Point | Mark | Difficulty | Session | Primary form | Objective demand                                             |
|-------|------|------------|---------|--------------|--------------------------------------------------------------|
| Q4-P2 | 1    | Standard   | 4       | Comparison   | Nesterov/RM Sprop mechanism distinction                      |
| Q4-P3 | 1    | Standard   | 4       | Numerical    | Correct raw and corrected moments                            |
| Q4-P4 | 1    | Stretch    | 4       | Numerical    | Correct signed update, epsilon condition, correction purpose |
| Q5-P1 | 1    | Easy       | 5       | Comparison   | Validation/test role distinction                             |
| Q5-P2 | 1    | Standard   | 5       | Scenario     | Correct trend diagnosis and checkpoint                       |
| Q5-P3 | 1    | Standard   | 5       | Scenario     | Correct checkpoint-4 gap with units /interpretation          |
| Q5-P4 | 1    | Stretch    | 5       | Scenario     | Targeted response with valid limitation                      |
| Q6-P1 | 1    | Easy       | 6       | Numerical    | Correct effective kernels and substitution                   |
| Q6-P2 | 1    | Standard   | 6       | Numerical    | Correct output shape and parameter total                     |
| Q6-P3 | 1    | Standard   | 6       | Scenario     | Separate dilation/stride interpretation                      |

| Point | Mark | Difficulty | Session | Primary form | Objective demand                                         |
|-------|------|------------|---------|--------------|----------------------------------------------------------|
| Q6-P4 | 1    | Stretch    | 6       | Scenario     | Parameter-sharing interpretation and validity conditions |
| Q7-P1 | 1    | Easy       | 7       | Scenario     | Coherent staged freeze /fine-tune plan                   |
| Q7-P2 | 1    | Standard   | 7       | Scenario     | Domain mismatch and overfit check                        |
| Q7-P3 | 1    | Standard   | 7       | Scenario     | Correct residual projection shape/stride                 |
| Q8-P1 | 1    | Easy       | 8       | Property     | Capacity/optimization/generalization condition           |
| Q8-P2 | 1    | Standard   | 8       | Property     | Dropout limitation and split-based distinction           |
| Q8-P3 | 1    | Stretch    | 8       | Scenario     | Evidence-linked diagnosis and bounded experiment         |

## Exact blueprint audit

- Sessions: S1 4, S2 4, S3 4, S4 4, S5 4, S6 4, S7 3, S8 3 = 30.
- Forms: property 5, comparison 4, numerical 12, scenario 9 = 30.
- Numerical/applied: 12+9=21 of 30.
- Difficulty: Easy 9, Standard 15, Stretch 6.
- Largest-remainder record: 30 one-mark units times shares 0.30,0.50,0.20 gives quotas/floors 9,15,6, all remainders 0; no tie-break needed.
- Time guidance: 10+10+12+12+12+12+10+12=90 minutes.